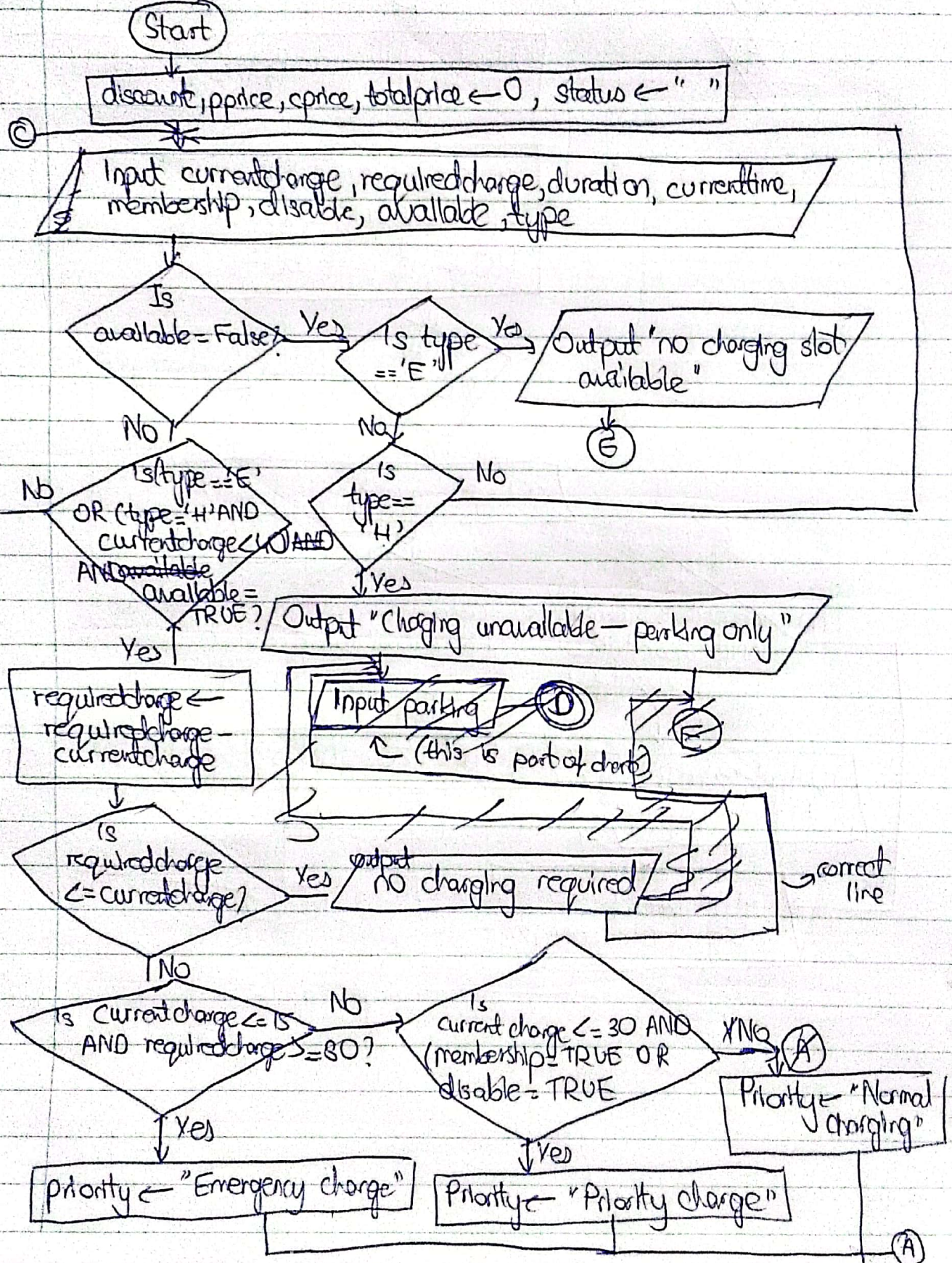
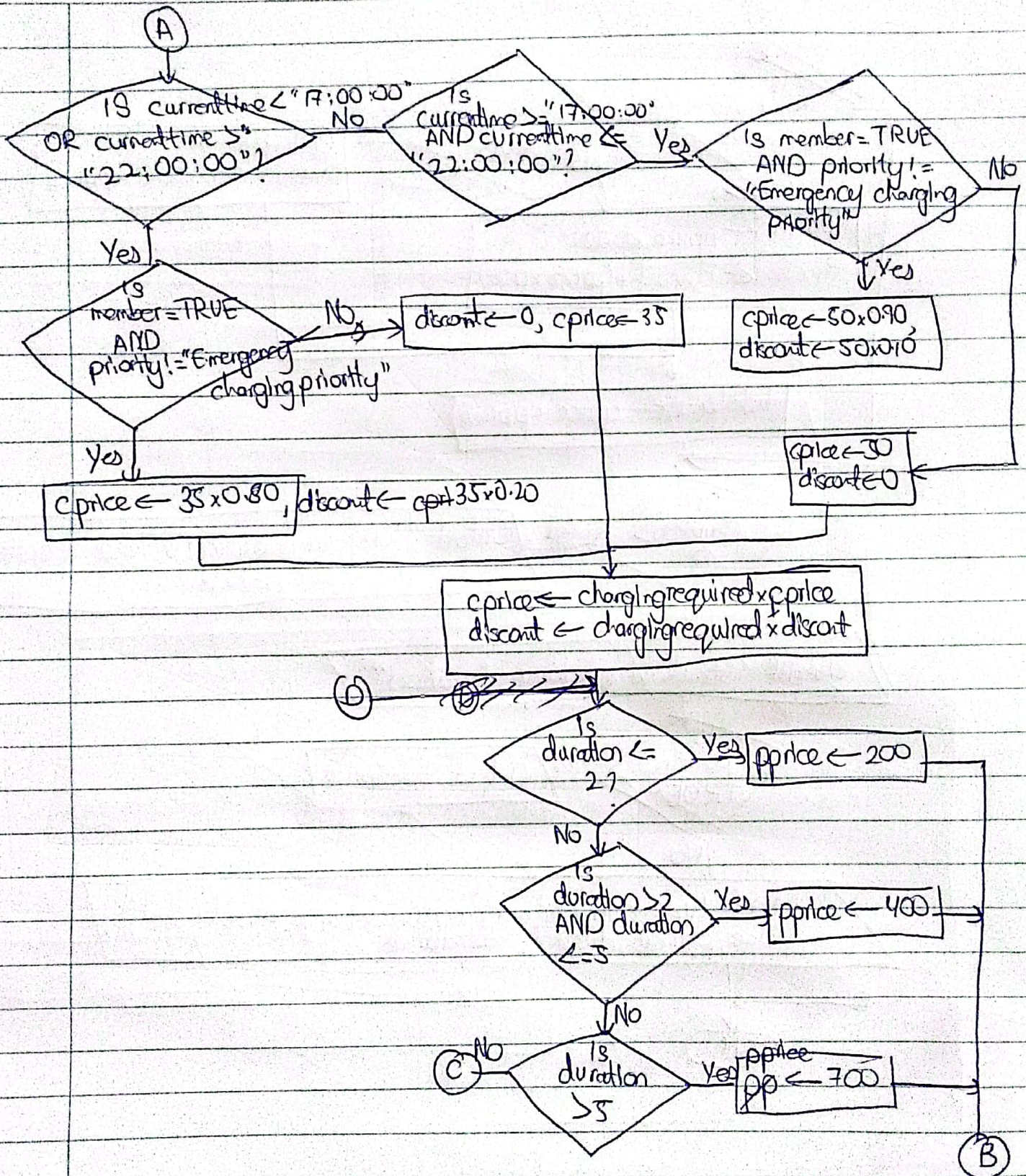
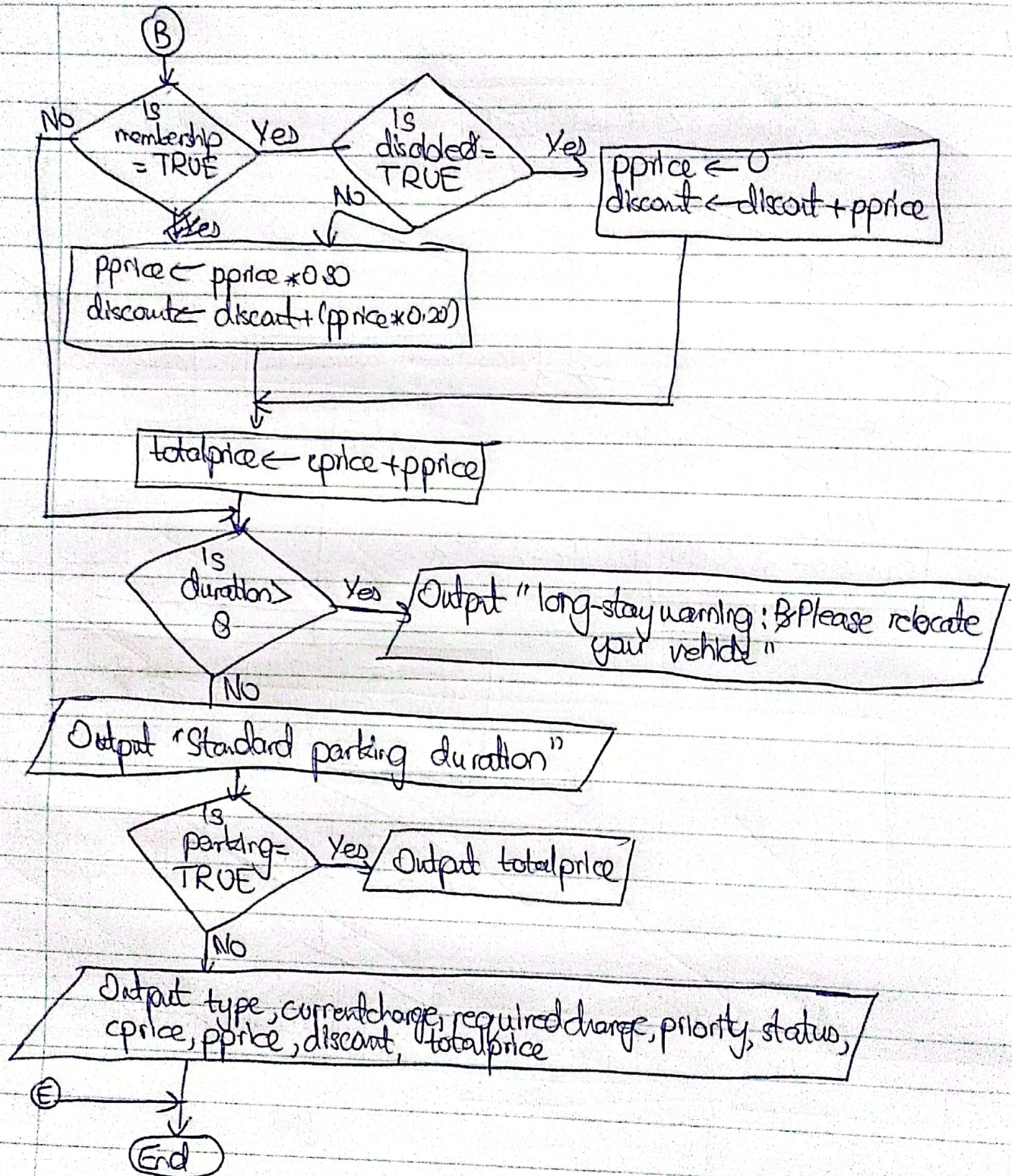


Q6: Flowchart:







Q6: Declare type: Char
 Declare currentcharge, requiredcharge, duration, discount: real
 Declare currenttime: time
 Declare membership, disable, available, parking: Boolean ^{charge}
 Declare ^{priority} priority, status: String, cprice, pprice, totalprice: INT Real
 discount, pprice, cprice, totalprice $\leftarrow 0$, status $\leftarrow$ "", parking, charge $\leftarrow$ FALSE
 Output "Enter current charge, requiredcharge, duration, currenttime"
 (in 24 hr clock), membership (Yes or No), Disable (Yes or No),
 available (Yes or No), type (E for EV, H for hybrid): "
 Input currentcharge, requiredcharge, duration, currenttime,
 membership, disable, available, type
 If available = False then
 If type = 'E' then
 Output "No charging Slot available" ~~break~~ break; stop
 Else if type = 'H' then
 Output "Charging unavailable - parking only"
 Endif ^{input parking}
 Endif ^{AND(available=TRUE)}
 If ((type='E') OR (type='H' AND currentcharge < 40)) then
 requiredcharge $\leftarrow$ requiredcharge - currentcharge
 If requiredcharge <= currentcharge then
 Else Output "no charging required"
 Else if (currentcharge <= 15) AND (required charge >= 80) then
 priority $\leftarrow$ "Emergency Charging Priority"
 Else if (membership = TRUE OR disable = TRUE) AND
 currentcharge <= 30 then
 priority $\leftarrow$ "Priority charging"
 Else

this line pay
offer else
at the
error

priority $\leftarrow$ "Normal charging"

EndIf

EndIf

charge $\leftarrow$ True

EndIf

If currenttime $<$ "17:00:00" OR currenttime $>$ "22:00:00" then
status $\leftarrow$ "off"

If member = ~~TRUE~~ AND priority $\neq$ "Emergency charging Priority" then

cprice $\leftarrow$ 35 $\times$ 0.80 discount $\leftarrow$ $\frac{cprice \times 0.20}{35}$

Else

cprice $\leftarrow$ 35

discount $\leftarrow$ 0

EndIf

Else If currenttime $\geq$ "17:00:00" AND currenttime $\leq$ "22:00:00"

then

status $\leftarrow$ "peak" If member = ~~TRUE~~ AND priority $\neq$ "Emergency charging Priority" then

cprice $\leftarrow$ 50 $\times$ 0.90 discount $\leftarrow$ $\frac{cprice \times 0.10}{50}$

Else

cprice $\leftarrow$ 50, discount $\leftarrow$ 0

EndIf

EndIf

EndIf

~~charge~~ $\leftarrow$ ~~TRUE~~ ~~charge~~ $\leftarrow$ ~~TRUE~~

cprice = ~~charging required~~ $\times$ cprice, discount $\leftarrow$ discount $\times$ ~~charging required~~

If parking = TRUE
OR charge =

If duration $\leq$ 2 then

pprice $\leftarrow$ 200

Else if duration $>$ 2 AND duration $\leq$ 5 then

pprice $\leftarrow$ 400

Else if duration $>$ 5 then

pprice $\leftarrow$ 700

EndIf

EndIf

Output "Vehicle does not
qualify for EV
charging"

EndIf

EndIf

If membership = TRUE then

pprice $\leftarrow$ pprice * 0.80

discount $\leftarrow$ discount + (pprice * 0.20)

Else if disable = TRUE then

pprice $\leftarrow$ 0

discount $\leftarrow$ discount + pprice

EndIf

EndIf

EndIf ~~EndIf~~

totalprice $\leftarrow$ cprice + pprice

If duration > 8 then

Output "long-stay warning: please relocate your vehicle after charging"

Else

Output "Standard & parking duration"

EndIf

If parking = TRUE then

Output ("Parking price: ", totalprice)

Else

Output type, currentcharge, requiredcharge, priority, status, cprice, pprice, discount, totalprice.